

Ridge Regularization

Ridge regression (L2 regularization) is a technique used to improve linear regression models by reducing overfitting and handling multicollinearity.

It works by adding a penalty proportional to square of magnitude of coeff (L2 norm) to the loss function, shrinking estimates toward zero to increase model stability.

How it works: It minimizes the residual sum of square (RSS) plus penalty term.

$$RSS + \lambda \sum \beta^2 \Rightarrow \sum (y_i - \hat{y}_i)^2 + \lambda (m)^2$$

The penalty (λ): The hyperparameter λ controls the strength of the penalty. If $\lambda = 0$, it is ordinary least square (OLS)

$\lambda = \infty$, the coefficients approach zero.

Multicollinearity solution: When predicted variables are highly correlated (multicollinearity). OLS estimates become unstable.

Ridge Regression reduces this variance by introducing bias, leading to more reliable predictions on new data.

key benefits: Improves model generalization & reduces sensitivity

to noise in the training data.

Disadvantages: It does not set coeff exactly to zero, means it does not perform feature selection and it introduces bias into the estimates.

Normalization: It is crucial to standardize independent variable before applying ridge regression so that the penalty is applied equally to all Features.